

What We Tested & Why

Goal

Improve AI Assistant response quality and consistency to drive higher user satisfaction scores without infrastructure changes.

Hypothesis

If we use a **structured chain-of-thought prompt** (Variant C), then satisfaction will rise because the AI logic will be more transparent and aligned.

Variants

A (Control): Baseline system prompt.

C (Treatment): Structured prompt with hidden chain-of-thought.

Primary KPI

Satisfaction Rating: Percentage of interactions receiving a 4 or 5-star rating from the user.

| What We Found

Variant A (Control)

59.0%

Variant C (Treatment)

69.15%

+17.21% Relative Uplift

Significance: 95% Confidence ($p < 0.0001$)

Z-Score: 5.40 | **CI:** 6.49% to 13.82%

*Note: Methodology followed strict stopping conditions (N=1,300) to avoid false-positives from early peeking.

Recommendation

Strategic Verdict: Roll Out

The structured prompt (Variant C) demonstrated clear statistical superiority. The improvement in user perception is both stable and significant.

Next Steps:

1. Permanent deployment of Variant C.
2. Decommissioning of legacy prompt A.

Risks & Mitigation

- **Latency:** CoT reasoning adds token generation time. Monitor secondary bounce rates closely.
- **Edge Cases:** Conduct 14-day observation for potential hallucinations in niche topics.
- **Iteration:** Test further prompt simplification for cost reduction next quarter.

| Image Sources

 Thumbnail for https://cdn.prod.website-files.com/67af3db5bb3b892e61258bd4/684150a03f50d03f6f023a30_analytics-dashboard.png
Source: www.domo.com
